

本版改动说明（LaTeX 版 · 正文 10 页）

第一轮：一至五节（10 月 7 日）。第二轮：第六节（10 月 8 日），并更新了第四、五节。

10 页

正文 + GenAI statement;
References 在第 11 页

14 → 10

V4 用同样排版需要 14 页

94 条

正文删改（只删不写）

2 个 cell

code.ipynb 改动，只影响 2 张图

一、代码 · 提交版/code.ipynb

- 只改了画图的两个 cell：图按报告里的实际尺寸出图（约 2.9 英寸宽，300 dpi），插入报告时不再缩放，图内文字印出来是 6–7pt。原来是 7.5 英寸宽的图缩到报告里，字只剩约 3.5pt。
- 去掉图内标题（报告图注已写），图例去边框，x 轴第二行改为样本数。
- 重新运行整个 notebook：104 秒，0 报错。outputs/ 中只有 `fig_hlc_vs_y.png` 和 `fig_pca_clusters.png` 变化，其余所有表格和数值文件与上一版逐字节一致。改前备份未覆盖原结果。

Cell 30 · 3.2 相关性分析 · Figure 1（host_listings_count 分段的超级房东率柱状图）→ outputs/fig_hlc_vs_y.png

```
-# Figure drawing
-figure, axis = plt.subplots(figsize=(7.5, 4.5))
-axis.bar(range(len(host_size_bands)), host_size_bands["rate"], edgecolor="black", linewidth=0.6)
-axis.axhline(overall_superhost_rate, color="red", label=f"Overall {overall_superhost_rate:.1%}")
-for bar_position, band_rate in enumerate(host_size_bands["rate"]):
-    axis.text(bar_position, band_rate + 0.012, f"{band_rate:.1%}", ha="center", fontsize=9)
-axis.set_xticks(range(len(host_size_bands)))
-# Readable band names for the bins (0,1], (1,2], (2,5], (5,14], (14,50], (50,100], (100,700]
-BAND_LABELS = ["0-1", "2", "3-5", "6-14", "15-50", "51-100", "101+"]
-assert len(BAND_LABELS) == len(host_size_bands)
-axis.set_xticklabels([f"{band}\nn={count:," for band, count in zip(BAND_LABELS, host_size_bands["n"])]], fontsize=8)
-axis.set(ylim=(0, 0.55), xlabel="host_listings_count (listings per host)", ylabel="Superhost rate", title=f"Superhost rate by host_listings_count band (n={len(listings):,} reviewed listings)")
-axis.legend(loc="upper right")
-axis.spines[["top", "right"]].set_visible(False)
-plt.tight_layout()
-plt.savefig(OUTPUT_DIR / "fig_hlc_vs_y.png", dpi=150)
-plt.close.figure)

+# Figure drawing, sized for the report: 2.9 x 1.95 inches inserted at 100%, so the
+# 6-7 pt labels print at that size (the caption carries the title)
+with plt.rc_context({"font.size": 6.5, "axes.labelsize": 6.5, "xtick.labelsize": 6, "ytick.labelsize": 6}):
+    figure, axis = plt.subplots(figsize=(2.9, 1.95))
+    axis.bar(range(len(host_size_bands)), host_size_bands["rate"], edgecolor="black", linewidth=0.4)
+    axis.axhline(overall_superhost_rate, color="red", linewidth=0.8, label=f"Overall {overall_superhost_rate:.1%}")
+    for bar_position, band_rate in enumerate(host_size_bands["rate"]):
+        axis.text(bar_position, band_rate + 0.012, f"{band_rate:.1%}", ha="center", fontsize=6)
+    axis.set_xticks(range(len(host_size_bands)))
+    # Readable band names for the bins (0,1], (1,2], (2,5], (5,14], (14,50], (50,100], (100,700]
+    BAND_LABELS = ["0-1", "2", "3-5", "6-14", "15-50", "51-100", "101+"]
+    assert len(BAND_LABELS) == len(host_size_bands)
+    axis.set_xticklabels([f"{band}\nn={count:," for band, count in zip(BAND_LABELS, host_size_bands["n"])]])
+    axis.set(ylim=(0, 0.55), xlabel="host_listings_count band (second line: listings in band)", ylabel="Superhost rate")
+    axis.legend(loc="upper right", fontsize=6, frameon=False)
+    axis.spines[["top", "right"]].set_visible(False)
+    plt.tight_layout(pad=0.3)
+    plt.savefig(OUTPUT_DIR / "fig_hlc_vs_y.png", dpi=300)
+    plt.close.figure)
```

Cell 87 · 3.5 PCA 与聚类 · Figure 2（PCA 散点图 + K-Means 质心）→ outputs/fig_pca_clusters.png

```
-figure, axis = plt.subplots(figsize=(7.5, 5.5))
-sns.scatterplot(data=scatter_frame, x="PC1", y="PC2", hue="Status", palette=STATUS_COLOURS, s=8, alpha=0.3, linewidth=0, legend=False, ax=axis)
-sns.scatterplot(x=centres_in_pc_space[:, 0], y=centres_in_pc_space[:, 1], marker="X", s=90, color="green", ax=axis)
-# Number the centroids as in the report (cluster label + 1)
-for cluster_label, (pc1_score, pc2_score) in enumerate(centres_in_pc_space[:, :2]):
-    axis.annotate(str(cluster_label + 1), (pc1_score, pc2_score), xytext=(6, 6), textcoords="offset points",
-        fontsize=10, fontweight="bold", color="darkgreen")
-axis.set(
-    xlabel=f"PC1 ({explained_variance[0] * 100:.2f}% of variance)",
-    ylabel=f"PC2 ({explained_variance[1] * 100:.2f}% of variance)",
-    title=f"PCA of reviewed listings by superhost status (random sample of {len(scatter_frame):,} of {len(listings_reviewed):,})",
-)
-axis.legend()
```

```
- handles=[Patch(color=colour, label=status) for status, colour in STATUS_COLOURS.items()] + [Patch(color="green",
label=f"K-Means centroids (k={CHOSEN_K})")],
- loc="upper right",
-)
-plt.savefig(OUTPUT_DIR / "fig_pca_clusters.png", dpi=150)
-plt.close(figure)
+# Sized for the report: 2.95 x 2.25 inches inserted at 100% (the caption carries the title)
+with plt.rc_context({"font.size": 6.5, "axes.labelsize": 6.5, "xtick.labelsize": 6, "ytick.labelsize": 6}):
+    figure, axis = plt.subplots(figsize=(2.95, 2.25))
+    sns.scatterplot(data=scatter_frame, x="PC1", y="PC2", hue="Status", palette=STATUS_COLOURS, s=2, alpha=0.35,
linewidth=0, legend=False, ax=axis)
+    sns.scatterplot(x=centres_in_pc_space[:, 0], y=centres_in_pc_space[:, 1], marker="X", s=28, color="green", ax=axis)
+    # Number the centroids as in the report (cluster label + 1)
+    for cluster_label, (pc1_score, pc2_score) in enumerate(centres_in_pc_space[:, :2]):
+        axis.annotate(str(cluster_label + 1), (pc1_score, pc2_score), xytext=(3, 3), textcoords="offset points",
+        fontsize=7, fontweight="bold", color="darkgreen")
+    axis.set(xlabel=f"PC1 ({explained_variance[0] * 100:.2f}% of variance)", ylabel=f"PC2 ({explained_variance[1] *
+100:.2f}% of variance)")
+    axis.legend(
+        handles=[Patch(color=colour, label=status) for status, colour in STATUS_COLOURS.items()] +
+        [Patch(color="green", label=f"K-Means centroids (k={CHOSEN_K})")],
+        loc="upper right", fontsize=5.5, frameon=False,
+    )
+    plt.tight_layout(pad=0.3)
+    plt.savefig(OUTPUT_DIR / "fig_pca_clusters.png", dpi=300)
+    plt.close(figure)
```

二、排版 · Word → LaTeX

版式（与 GNN arXiv v1 同一套 elsarticle 样式，满足 spec）

- elsarticle [final,1p,11pt]：A4、单栏、11pt、单倍行距、页边距 2.54 cm。正文字体 TeX Gyre Termes（Times 风格），数学字体 STIX Two Math。
- 标题区：主标题 + 副标题，下一行“Group W06G02 · COMP20008 Elements of Data Processing, Assignment 2”，再下一行四位组员姓名和学号（spec 要求组号和姓名紧跟标题）。
- 节标题粗体，小节标题斜体；段首加粗的小标题统一改为斜体内联标题（论文常见写法）。
- 页脚只有页码。超长列名可以在下划线处断行，避免整行被撑开。
- 为挤进 10 页收紧了间距：节和小节标题上下留白、表格行高 0.98、表格和图与正文之间的距离、图注间距。字号、行距和页边距都没有动。

表格（数值逐字照抄 Word，只改排版）

- 全部改为三线表（booktabs），footnotesize 字号，偶数行加浅灰底纹。数值列右对齐，长表头分两行。
- Table 5（超参数 CV F1）：改为横排，k-NN 和决策树各占一块（取值 / 平均 F1 / ±SD 三行），选中的 k=101 和 max_depth=10 用浅蓝底加粗标出。
- Table 7（测试集表现）：Accuracy 每个模型只显示一次（Word 原表在两个类别行里重复）。
- Table 11（聚类画像）：K-Means 的 K1-K5 和 Ward 的 W1-W5 合并成一张表，表头分行，列宽逐列手调，避免溢出。
- Table 5 下方的脚注“† None means unlimited depth.”已删（第 2.3 节已有说明）。

图

- Figure 1 和 Figure 2 直接使用 code.ipynb 新生成的 300 dpi 图，按原尺寸插入（宽度分别为版心的 0.46 和 0.47）。
- Word 稿里的缩图操作（4 条）和“表格转图片”方案（11 条）都只在 Word 稿中做过，LaTeX 版没有用，因此不计入下面的删改清单。

结尾部分

- Generative AI statement 放在 Conclusion 之后，仍在第 10 页内；References 从第 11 页开始（spec：10 页不含 References）。
- GenAI statement 原文开头重复的“Generative AI statement:”已去掉，因为上方已有同名节标题。

三、论文内容 · 相对 V4 的全部删改

以下按类型列出，编号对应删改日志 all_log.tsv 中的行号。所有改动都是删除或调整结构，没有 AI 新写的句子（spec 8.4）。图表编号已随删除同步更新：V4 的 Figure 2 改为 Figure 1，Figure 5 改为 Figure 2；Table 11 改为 Table 10；Table 12 与 Table 13 合并为 Table 11。

删除句子/短语 50 条

#	原文 / 改动	理由
2	Performance over the preceding twelve months is assessed every three months, during a 7-day period that begins on the 1st of January, April, July or October, and hosts are notified of the result at the end of that period	评估周期细节与 RQ 无关

3	(number_of_reviews, number_of_reviews_ltm, reviews_per_month)	列名在表 9/特征说明中已出现
4	(host_response_rate, host_response_time, host_acceptance_rate)	同上
5	Preprocessing keeps the full sample and prepares both families of variables, correlation compares the strength of each family's links with superhost status, the supervised models and feature experiments compare how well each family predicts that status, feature selection identifies which variables carry the signal, and PCA and clustering show which family of features the underlying structure follows.	路线图, 章节标题已体现
7	This leads to deviation in class balance, thus the model would have an inflated sense of how common superhosts are.	前一句的数字已说明
9	When renting a private room, bedroom count is not relevant, so the host does not fill it in. Thus, imputation is not needed.	前后句已说明
10	; one-hot encoding all 82 categories would instead produce a large number of sparse columns	次要理由
14	, which is confirmed by an assertion in the code	实现细节
15	The target variable is y (binary).	y 已在 2.2 定义
17	With about 1,262 superhost listings in each fold, F1 is stable and the between-fold standard deviation is only around 0.01–0.02 (Table 5).	与上一句重复
18	, and the interval for DT – k-NN is [0.0130, 0.0986]	结果在 Table 8 和 4.3 节
19	The two groups remain unequal in size, with eight features against three, so part of the gap could reflect how much information a larger set can hold.	4.3 与第 5 章已写
20	rather than in a separate table	多余
21	, where the target y describes the clusters but is not used to form them,	次要
22	, used to check for degenerate clusters	次要
23	, and the drop of 8,780.1 from k=4 to k=5 almost halves to 4,346.8 at k=6	与百分比降幅重复
49	The 21,251 listings come from 11,474 hosts, so the observations are not independent.	第 5 章已写
50	The exploratory set is also uneven, with three operational variables against one fundamental. This imbalance is why the supervised stage ran the feature-subset experiments, where the two families are scored on equal terms	2.2 节已写
51	Both models perform much better on non-superhosts, where the decision tree reaches 0.845 precision.	Table 7 已列
52	, and the tree correspondingly has 530 false negatives against 694 for k-NN	上一段已写 530 / 694
53	One caveat is that the operational group has eight features and the fundamentals group has three, so part of the gap may simply reflect that a larger set can hold more information.	第 5 章已写
54	The rankings show where the two methods agree and where they diverge.	空泛
55	, which is expected given their strong association with the target	空泛
56	, making it one of 530 false negatives	3.4 节已写
57	Location is not absent, though, since dist_cbd dominates the second axis. The accurate claim is that the largest single axis is operational and the second is geographic.	PCA 段已写
58	K-Means shows two extreme clusters separating rather than all five differing: K3 at a 6.8% superhost rate and K2 at 53.9% stand apart, while the other three clusters, between 35.3% and 43.7%, stay close to the base rate.	K-Means 段已写
59	W1, the largest Ward cluster at 8,256 listings, draws from all five K-Means clusters, mainly K2 (4,410), K5 (1,813), K4 (1,099) and K3 (815).	细节, 交叉信息已足够
65	The CV F1 difference between depth 4 and depth 10 is only 0.0033, smaller than the standard deviation; under a one-standard-error rule, depth 4 would have been selected. Training F1 is 0.7188 against 0.6319 on the test set.	4.3 节已写

66	The clustering input was log-transformed on the three skewed features and then standardised, which gives every feature the same weight. That is what allows accommodates, which ranges from 1 to 16, to count as heavily as days_since_last_review, which ranges from 0 to 4,593	2.5 节已写
67	The linkages covered in the course collapse into one large cluster on this dataset (see the methodology), so only Ward is used in the comparison.	2.5 节已写
68	The report asks whether superhost status can be inferred from listing attributes and host operating behaviour once Airbnb's own criteria are excluded, and whether it is more closely associated with listing fundamentals or operational strategy.	复述 RQ
76	New amenity_count field derived from parsing the amenities field, with counts matching the JSON array length.	与 Table 1 和 2.1 重复
77	, while 29 rows genuinely have no amenities	次要
78	Because host_listings_count and availability_365 have many repeated values, they actually produce only 7 and 8 bins (see the Bins column in Table 4).	Table 4 的 Bins 列已显示
79	It rises from 28.5% for hosts with 0–1 listings to 46.1% for 3–5 listings, stays on a plateau between 40.1% and 42.7% for 6–100 listings, and then falls to 16.9% above 100.	4.2 节逐段解释时重复这些数字；Figure 1 图上有
80	days_since_last_review with dist_cbd gives Pearson -0.081 and Spearman $+0.061$, and host_listings_count with the target gives -0.069 and $+0.070$.	下面两段分别给出这两对的数字
81	The tree finds 1,047 of the 1,577 test superhost listings and misses 530, a recall of 0.664, while k-NN misses 694.	Table 7 与下一段已有
82	Their true prices may be higher or lower, and a median cannot recover them, while price_missing records that a value was imputed.	前一句已说明
83	Set F, with six features, only partly addresses this.	次要
84	Preprocessing shows what is at stake, since the 4,477 listings without a review carry a 6.59% superhost rate against 35.72% for reviewed listings.	结论只保留回答 RQ 的证据
85	Correlation points the same way: distance from the CBD is the weakest of the ten pairs at NMI 0.0047, while review recency reaches 0.0785 against the target	同上
87	Retraining on only the top three features confirms their value. The filter top three give CV/test F1 of 0.5820/0.6331 for the decision tree and 0.5662/0.5865 for k-NN; the embedded top three give 0.5628/0.6250 and 0.5525/0.5635, compared with 0.5745/0.6319 and 0.5370/0.5773 for all eleven features. Three operational features therefore match or exceed the full model for the decision tree.	同上
102	This report asks what else, besides the criteria themselves, separates superhosts from other hosts in the Melbourne market.	RQ 已在上方陈述
103	Calculating amenity_count was selected as amenity_count separates superhosts at 44.08 amenities against 31.03 amenities.	数字已在 Table 1
104	Without standardisation the availability feature (0 to 365) would outweigh sleeping capacity (1 to 16).	标准化的必要性第 5 章已论述
105	(from the dataset's id column)	多余说明
106	PCA was run on the standardised six-feature set.	2.5 节已写
107	Using superhost F1 scores, we can conclude that superhost status can be partly inferred from the listing record once the criteria are removed.	结论在第 6 章
108	Activity is associated with the clearest separation, but location also distinguishes the clusters: K1 is 1.42 km from the CBD while K4 and K5 are 20.56 km and 17.82 km away, even though their superhost rates lie only between 35.3% and 43.7%.	K-Means 段已给出 K1/K4/K5 的距离和超级房东率
109	Superhosts are 30.65% of listings, so accuracy is a poor guide and a majority-class classifier already reaches 0.694.	2.3 与 4.3 已写

删除段落 17 条

#	原文 / 改动	理由
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25	(图段落)	SSE 降幅与 silhouette 数字已在正文
26	Figure 1: Elbow curve of K-Means SSE for k=2-8 for S3. Dotted line indicates chosen hyperparameter k=5.	SSE 降幅与 silhouette 数字已在正文
27	(图段落)	TP/FP/FN 数字已在 4.3 正文
28	Figure 3: Confusion matrices for both models on the test set (n=5,146). Of the 1,577 test superhost listings, k-NN finds	TP/FP/FN 数字已在 4.3 正文
29	(图段落)	置换重要性数值已在 Table 9
30	Figure 4: Permutation importance for the decision tree (left) and k-NN (right) across all 11 features, coloured by famil	置换重要性数值已在 Table 9
33	The measurable before/after impact of our 3 chosen preprocessing steps is presented in Table 3.	表格标题已说明
34	Table 4 reports Pearson, Spearman, MI and NMI for all 10 pairs.	表格标题已说明
35	5-fold stratified, host-grouped cross-validation F1 scores per hyperparameter value are reported in Table 5.	表格标题已说明
36	Cross-validation results compared to default hyperparameters are reported in Table 6.	表格标题已说明
37	Test-set performance is reported in Table 7.	表格标题已说明
38	Table 8 gives the 95% bootstrap interval for each model, for the gap between them, and for the feature-subset comparison	表格标题已说明
39	Cluster profiles. Tables 12 and 13 give the size, superhost rate and median feature values of every cluster under each m	表格标题已说明
40	† None means unlimited depth.	2.3 节已说明 None 的含义
44	Table 10: Hard-case listing 6812677 compared against class medians from the 20,582 training listings.	属性值已在 4.4 正文逐项引用
45	Table 13: Hierarchical (Ward) cluster profiles at the same k=5. Feature values are medians, in original units (km, guest	与 Table 12 合并
86	Top 3 features: The top three features from each method are validated by retraining both models on those three and compa	额外分析, rubric 未要求; 讨论段同步删除

小标题并入段落 12 条

#	原文 / 改动	理由
1	Research question:	标题行单独占一行
6	Justifications for selections:	小标题单独占行
8	Justifications for rejections:	小标题单独占行
11	Variable set:	小标题单独占行
12	Rationale behind variable set:	小标题单独占行
13	Computation:	小标题单独占行
16	Test/train split:	小标题单独占行
60	Preprocessing	第 5 章 5 个三级标题各占一行
61	Correlation	第 5 章 5 个三级标题各占一行
62	Supervised Learning	第 5 章 5 个三级标题各占一行
63	Feature Selection	第 5 章 5 个三级标题各占一行
64	PCA & Clustering	第 5 章 5 个三级标题各占一行

替换 6 条

均为删去一段后的衔接改动（删去失效的图表引用、合并后的表题），没有新写内容。

#	原文 / 改动	理由
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24	The hyperparameter k for K-Means is chosen from the elbow curve over k=2–8 (Figure 1). → (删去)	删除 Figure 1 后的引用
41	The rate climbs from 28.5% at one listing or fewer to 46.1% at three to five listings, then falls to 16.9% above 100. → (删去)	与正文上一段重复
42	, and Figure 4 compares the permutation importance of the two models across all 11 features → (删去)	Figure 4 已删除
43	, and Table 10 compares its attributes against the median of each class → (删去)	Table 10 已删除
46	K-Means cluster profiles (k=5) → K-Means (K1–K5) and Ward (W1–W5) cluster profiles (k=5)	合并后的标题
48	Figure 4 → Table 9	Figure 4 已删除

合并表格 1 条

#	原文 / 改动	理由
47	Table 13 的 5 行 Ward 并入 Table 12	省一个表头和标题

合并单元格行 1 条

#	原文 / 改动	理由
75	四位组员姓名合为一行	省 3 行

重新编号 5 条

#	原文 / 改动	理由
70	Figure 2 → Figure 1	2 处
71	Figure 5 → Figure 2	1 处
72	Table 11 → Table 10	2 处
73	Table 12 → Table 11	2 处
74	Figure 5 → Figure 2 (图注跨 run)	上一轮遗漏

删除空段落 1 条

#	原文 / 改动	理由
69	8 个	空行占版面

移动段落 1 条

#	原文 / 改动	理由
88	Generative AI statement 先移到参考文献之后 (Word 稿), LaTeX 版再移回 Conclusion 之后、References 之前	GenAI statement 留在 10 页内, References 单独成页

四、文件位置

文件	说明
report/LaTeX版(10页)/A2 Report (LaTeX版).pdf	最终报告
report/LaTeX版(10页)/report.tex + figures/	报告源文件, 以后直接改它再编译
report/LaTeX版(10页)/构建脚本/	从 Word 稿重新生成 report.tex 的脚本和版式模板
report/LaTeX版(10页)/数字溯源表.pdf	报告和 PPT 中每个数字的来源 (第二轮新增)
Project 2 代码/GitHub修正版/提交版/code.ipynb + README.txt + outputs/	代码、README 和全部输出
PPT: claude.ai/artifact/JhZUNWLV2eZa1YchD4JckJ	19 页, 仅本人可见, 需在 Share 里共享给组员

五、提交前还没做的

- **GenAI statement 必须改**：现在写的是 Claude “solely to review drafts...”，与实际不符（还做了篇幅删减、LaTeX 排版、重画图 and slides）。spec 8.4 要求如实声明，需你们自己写。补完后重新编译，确认仍在第 10 页内。
- **README.txt 里引用的图表编号改成 Figure 1–2、Table 1–11**——已在第二轮完成。
- 可选：Table 3、6、7、8 在正文中未被引用（引出句在压缩时删掉了），如需可自行补“(Table 7)”之类的引用。

六、第二轮迭代（10 月 8 日）

1. README.txt

- 第 4 节对照表按最终报告重写：每一项写明小节号、Table / Figure 编号（Figure 1–2、Table 1–11）、输出文件（JSON 键名）和 notebook 节号；逐项对照 report.tex 核对过位置。
- 报告未展示但 notebook 仍生成的文件单独列出：fig_elbow.png、fig_confusion.png、fig_importance.png、fs_validation.csv、fig_dendrogram.png。
- 改正一处不准确的说法：原文称“报告的每个数字都在 4 个 evidence_*.json 中”，实际也有 CSV 和图，改为“在下表列出的文件中”。
- 新增一行：2.3 节 price 与 accommodates 的 Spearman 0.682 → evidence_preprocess.json (price_accommodates_spearman)，notebook 1.6b。
- MI 分箱敏感性一行注明：JSON 中为 nats，报告引用的是 bits (= nats ÷ ln 2)。

2. code.ipynb

- 报告 2.3 节引用的 0.682 原先没有任何代码输出支持。1.6b 节新增两项证据（有标价的 19,175 条房源上的 Spearman = 0.6821）：

Cell 18 · 1.6b Further evidence cited in the report

```
+ # Price rises with capacity, the argument for price sitting near the fundamentals
+ # (report Section 2.3); measured on the listings whose price is observed, not imputed
+ "price_accommodates_spearman": round_value(listings.loc[~price_is_missing,
"price_num"].corr(listings.loc[~price_is_missing, "accommodates"], method="spearman")),
+ "price_accommodates_spearman_n": int((~price_is_missing).sum()),
- "largest_host_listing_count_in_data", "superhost_share_if_price_missing_dropped",
"other_room_type_count"]}]
+ "largest_host_listing_count_in_data", "superhost_share_if_price_missing_dropped",
"other_room_type_count",
+ "price_accommodates_spearman", "price_accommodates_spearman_n"]}]
```

- 重新运行：0 报错；outputs/ 中只有 evidence_preprocess.json 变化（新增上面两个键，原有键值全部不变）。
- 从零复现检查：在临时目录只放 code.ipynb 和原始 listings.csv 运行，111 秒完成，19 个输出文件与提交版逐字节一致。
- 静态检查：90 个单元格按顺序执行；无绝对路径、TODO、中文、调试残留；注释约占代码行 23%；README 中的软件版本与运行环境一致。

3. 报告

- 正文没有改动。检查结果：A4、单栏、11pt；正文 + GenAI statement 在 10 页内，References 在第 11 页；第一页有组号和四位组员。
- 新增 **数字溯源表.pdf** (36 页)：报告 720 个数字（输出 635 · 推导 38 · 设定 47）、PPT 332 个数字（输出 275 · 推导 37 · 设定 20）逐个列出上下文和来源，**没有找不到来源的数字**。推导值均写明算式，例如 $0.4434 = 0.6389 - 0.1955$ 、 $99.2\% = 3,224 \div 3,249$ 、 $MI\ 0.1695 = 0.1175\ nats \div \ln 2$ 。

4. PPT（16 页 → 19 页）

- 动画：每页主要内容改为可动画的固定定位元素，加入逐根升起的柱状图、从左划入的条形、依次弹出的卡片；章节末页用推入切换；关键结论（如第 12 页 0.639）点击后出现，演讲备注里用 [click] 标出。动画仅在演示模式播放。
- 新图 5 张：PCA 散点（编号质心）、两个模型的混淆矩阵、肘部曲线 + silhouette、Ward 树状图（5 簇着色）、评论间隔随距离变化曲线。均在临时目录复制 notebook 运行后按 PPT 配色重画，数字与报告一致；提交版 notebook 和 outputs 未改动。
- 新增 3 页：第 8 页“距离的符号反转”（Pearson -0.081 / Spearman +0.061，报告 4.2）、第 11 页“混淆矩阵”（1,047 对 883，690 对 599）、第 16 页“选择 $k = 5$ 与 Ward”。
- 按最终报告改内容：features 页“Filter top 3: 0.582”（对应实验已从报告删除）换成实验 E（去掉 availability_365: CV 0.5846 对 0.5745）；criteria 页出处改为参考文献标题“How to become a Superhost”。
- 封面：两个圆环换成由 389 个点拼成的房子（每点约 66 条房源），动画中 119 个点（30.65%）亮成橙色代表超级房东房源。